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PRESENTATION FINAL PROJECT – UNIT 2

EIGENVALUES & EIGENVECTORS IN AI

Abdulmajeed Alshehri

Abdullah Alkhurayjah

Mohamed Alsomali

Saad Alshahrani

Nawaf Alorabi

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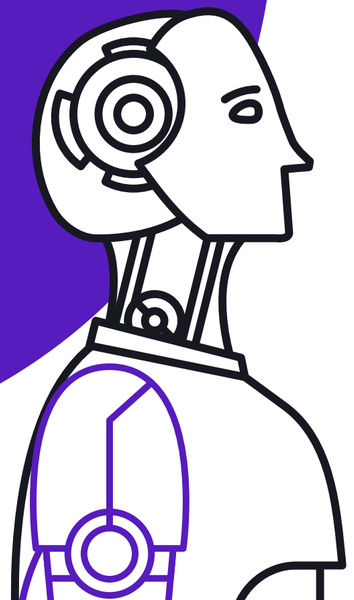

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TOOLS USED

NumPy

pandas

scikit learn
machine learning in Python

matplotlib

CO

kaggle

SymPy

GitHub

PROBLEM STATEMENT

In Artificial Intelligence, we often deal with high-dimensional data.

More dimensions mean:

01 More complexity

02 More noise

03 Higher computational cost

ABOUT THE DATASET

This dataset contains information about Spotify tracks across 125 different music genres,

df.shape
(114000,21)

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WHY THIS MATTERS IN DATA

01

IS THERE ANY SPREAD?

Data spreads differently across directions

02

IS THERE ANY INFO?

Some directions contain more information.

03

WHICH PCS MOSTLY CONTAIN NOISE?

SOME DIRECTIONS MOSTLY CONTAIN NOISE.

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WHAT IS AN EIGENVALUE?

An eigenvalue (λ) tells us:

How much scaling happens in that direction

Large λ → strong stretching

Small λ → weak effect

WHAT IS AN EIGENVECTOR?

$$A\vec{x} = \lambda\vec{x}$$

$n \times n$
Matrix

Eigenvector

Eigenvalue

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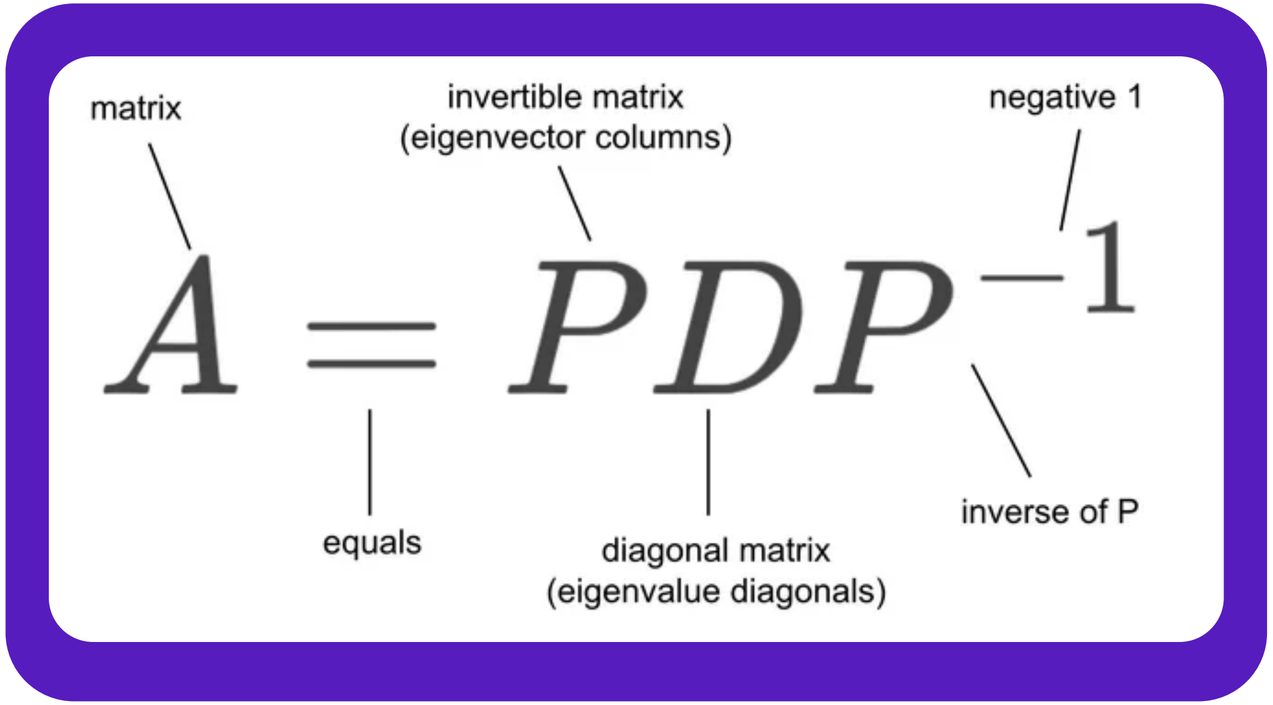
DIAGONALIZATION

Diagonalization breaks a matrix into directions (P), their importance (D), and transforms back to the original space (P^{-1}), making complex data easier to understand."



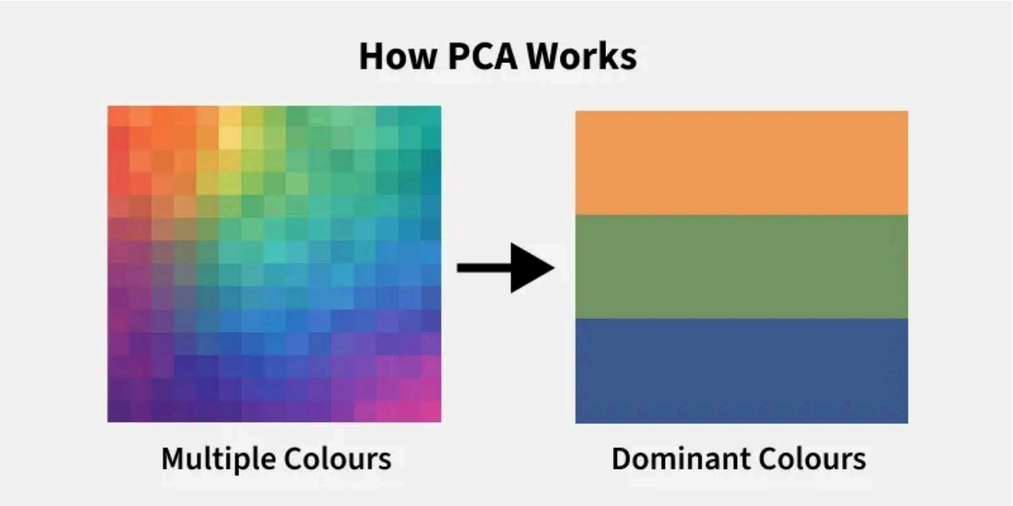
CONNECTION TO AI

- Feature reduction
- Image compression
- Face recognition
- Recommender systems

WHAT IS PCA?

Principal Component Analysis (PCA) is:

- Reduces dimensionality of data, removes noise, and extracts key patterns.
- Common in image recognition, machine learning, and data compression.



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HERE'S HOW IT WORKS:

- Step 1: Standardize the Data
- Step 2: Calculate the Covariance Matrix
- Step 3: Find the Principal Components
- Step 4: Reduce the Dimensions

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PRINCIPAL DIRECTIONS

Each Principal Component:

- Is an eigenvector of the covariance matrix
- Points to a direction of maximum variance

PC1 → MAXIMUM VARIANCE
PC2 → SECOND MAXIMUM
PC3 → THIRD MAXIMUM

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REFERENCES

Dataset

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CODE

GitHub

TUWAIQ-LINEAR-ALGEBRA

THANK YOU

NUMPY PROJECT